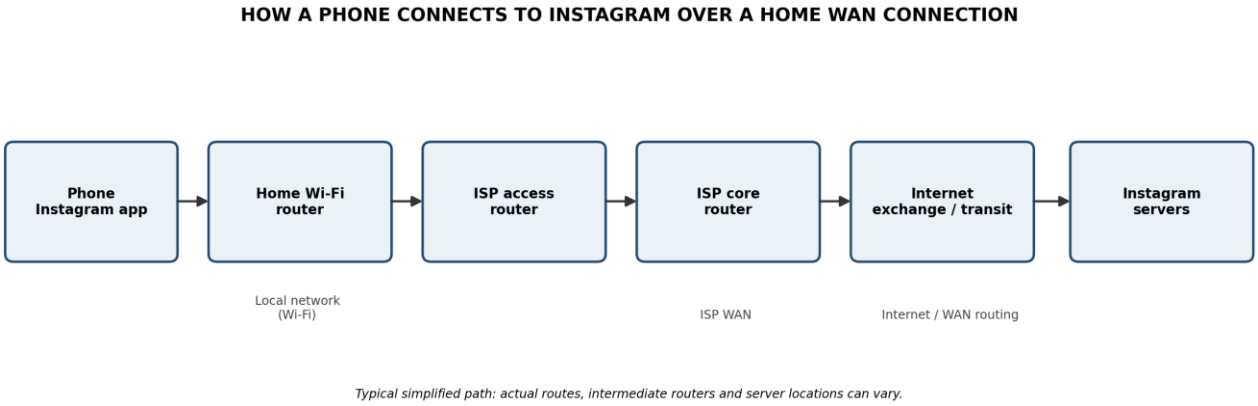
WIDE AREA NETWORKS (WAN), VPNs AND CONNECTIVITY

Questions and Answers

This assignment explains how a phone reaches an online service through a home Wi-Fi network and an Internet Service Provider (ISP), compares common WAN technologies, describes VPN tunnelling, and recommends a WAN solution for connecting multiple warehouses.

Question 1. Phone connection to Instagram through home Wi-Fi

When a person opens Instagram on a phone connected to home Wi-Fi, the phone first sends network traffic through the wireless router. The router forwards it to the ISP network. The traffic then travels through WAN routers and Internet transit or exchange networks until it reaches the data-centre network hosting Instagram's services. The response returns along an available network route.



Explanation of the devices in the path

- **Phone:** Runs the Instagram app and sends requests, such as loading a feed or viewing a photo.
- **Home Wi-Fi router:** Connects the phone to the home network and forwards traffic toward the ISP, commonly using network address translation (NAT).
- **ISP access router:** Connects the customer's broadband connection to the provider's network.
- **ISP core router:** Forwards traffic across the ISP's WAN toward other networks.
- **Internet exchange / transit network:** Allows networks to exchange traffic or carry it onward toward Instagram's network.
- **Instagram servers:** Receive the request and send the requested content back to the phone.

Note: This is a simplified learning diagram. Real Internet routing can involve additional routers, content delivery networks (CDNs), security systems, and different paths depending on location and network conditions.

Question 2. Comparison of three WAN technologies

A Wide Area Network (WAN) connects networks over large geographic areas. The table compares three common options. Actual speeds and prices depend on the provider, service-level agreement, location, and bandwidth selected.

WAN technology	Typical speed / performance	Relative cost	Real-life use
MPLS (Multiprotocol Label Switching)	Provider-managed service; can offer predictable performance and traffic prioritisation. Bandwidth is selected to suit the organisation.	High to medium-high; usually more expensive than ordinary broadband.	A large company connecting branch offices, warehouses, and data centres that need managed connectivity and quality of service.
Leased line	Dedicated point-to-point connection with committed bandwidth; often symmetrical and consistent.	High; the organisation pays for a dedicated circuit, especially over long distances.	A bank, large office, or data centre needing a reliable dedicated link between two sites.
Broadband Internet	Varies by technology and plan; fibre can provide high speeds, while performance may vary with congestion and service conditions.	Low to medium; generally cost-effective and widely available.	A home Internet connection, small office, or a branch using secure Internet-based services.

Summary: Broadband is usually the economical choice for homes and smaller sites. Leased lines provide a dedicated connection between locations. MPLS is useful when an organisation wants a provider-managed WAN that can connect many sites and prioritise important business traffic.

Question 3. How a VPN uses a WAN to provide secure access

A Virtual Private Network (VPN) creates an encrypted connection, often called a private tunnel, across an existing network such as the public Internet. The Internet is a WAN because it carries data between networks over large geographic areas. At a coffee shop, a phone or laptop connects to the public Wi-Fi router, and its Internet traffic travels through the coffee shop's ISP and the wider Internet.

When the user turns on a trusted VPN app, the device first establishes an authenticated connection to a VPN server. The VPN software encrypts the data before it travels across the public Wi-Fi and WAN. People sharing the Wi-Fi network or operating intermediate network equipment should not be able to read the protected contents of that traffic. The VPN server decrypts the traffic and forwards it to the destination service, such as Spotify or YouTube. Return traffic travels back through the VPN tunnel and is encrypted on the way to the device.

Simplified VPN path

1. Phone or laptop connects to the coffee shop's public Wi-Fi.
2. The device establishes an authenticated, encrypted tunnel to the VPN provider's server.
3. Encrypted VPN traffic crosses the coffee shop network, ISP, and public Internet (the WAN).
4. The VPN server forwards the request to Spotify or YouTube and returns the response through the tunnel.

A VPN helps protect data in transit on untrusted Wi-Fi and can hide the user's traffic from local network observers. It does not make a device completely anonymous or protect it from phishing, malware, unsafe downloads, or every tracking method. Users should still use HTTPS, keep devices updated, and choose a reputable VPN provider.

Question 4. Best WAN technology for Flipkart warehouses

For connecting Flipkart's warehouses in Ahmedabad, Mumbai, and Delhi, a suitable recommendation is an MPLS-based WAN, with secure Internet VPN links as backup where practical.

Reasons for the recommendation

5. **1. Reliable connectivity between multiple sites:** MPLS is designed for connecting geographically separated business locations through a provider-managed network. It can give the warehouses dependable access to central inventory, order-management, and logistics systems.
6. **2. Traffic prioritisation and managed performance:** The provider can configure quality-of-service policies so important applications, such as inventory updates, warehouse scanners, voice communications, and dispatch systems, receive suitable priority.

The final design should also consider cost, service availability at each warehouse, bandwidth requirements, security, and recovery arrangements. If cost is the main priority and each site has good fibre broadband, business-grade Internet connections with site-to-site IPsec VPNs may be a lower-cost alternative. For a large operation where predictable managed connectivity is important, MPLS is a strong choice.

Conclusion

WANs make it possible for phones, businesses, and online services to communicate across long distances. Different WAN technologies suit different needs: broadband is economical, leased lines provide dedicated connectivity, and MPLS supports managed multi-site networks. A VPN adds encryption and authentication over a public WAN, helping users protect their data when connecting through public Wi-Fi.